

A LOAN RECORDS DATA ANALYSIS PROJECT

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1. INTRODUCTION

1.1 PROJECT OVERVIEW

This project presents an analysis of loan data with the goal to uncover insights. The project transforms raw data using python and Power BI into information that could help in decision making in the lending industry by stakeholders and key decision makers.

1.2 OBJECTIVES

- To analyze loan distribution across listed employment types
- To classify borrowers based on income and credit score categories
- To identify trends in loan issuance
- To prepare clean and well structured datasets for Power BI reports
- Demonstrate technical proficiency in data analysis and visualization

1.3 DATA DESCRIPTION

The dataset contains 255,347 records. The key fields are:

LoanID: A unique identifier for each loan in the dataset

Age: The borrower's age at the time the loan was issued

Income: The borrower's annual income

LoanAmount: The total amount of the loan that the borrower is requesting or has been approved for

CreditScore: A numerical representation of the borrower's creditworthiness, typically ranging from 300 to 850. A higher credit score indicates the borrower is more likely to repay the debt

MonthsEmployed: The number of months the borrower has been employed their current job or with their current employer

NumCreditLines: The total number of active credit lines(e.g credit cards, loans) the borrower has at the time of the loan

IntrestRate: The annual percentage rate(APR) charged for borrowing the loan amount, usually expressed as a percentage

LoanTerm: The length of time(in months) over which the loan is to be paid

DTIRatio: The DTI ratio, which measures the borrower's debt payments relative to their income. A higher ration can indicate greater financial stress

Education: The highest level of education the borrower has completed(e.g High school, Bachelor, Masters)

EmploymentType: The type of employment the borrower is engaged in(E.g Full-Time, Part-Time, Self-employed)

Marital status: The marital status of the borrower(single, married or divorced)

HasMortgage: An indicator(Yes/No) that shows whether the borrower has an existing mortgage on a property

HasDependents: An indicator(Yes/No) that shows whether the borrower has dependents(children, other family members) to support

LoanPurpose: The primary reason for taking the loan(Home purchase, Debt consolidation, Education)

HasCoSigner: An indicator (Yes/No) that shows whether the borrower has a co-signer for the loan(someone who agrees to take responsibility if the borrower defaults)

Default: An indicator(Yes/No) that shows whether the borrower defaulted on the loan or failed to make timely payments.

Loan Date(DD/MM/YY): The date the loan was issued or originated

2. DATA PREPARATION

2.1 DATA CLEANING AND FORMATTING

The dataset was clean with fields complete, consistently formatted and free from duplicates and anomalies.

2.2 FEATURE ENGINEERING

To support creation of targeted analysis and visualization, new columns were created to highlight key patterns. The created features enabled clearer insights into borrower behavior, loan distribution and risk levels.

The columns:

IncomeCategory: Groups borrowers into Low, Middle or High income tiers

ScoreCategory: Classifies credit scores into very low, low, medium or high

LoanYear: Extracts the year from the date column

DefaultRateByIncome: shows the default rate for each category of income

AverageLoanByIncome: calculates the average loan amount per income group

LoanAmountByScoreGroup: aggregates loan amounts by credit score category

EmploymentLoanShare: gives the loan distribution across employment types

IncomeLoanAggregation: summarizes total loan amounts by income category

3. INCOME ANALYSIS

The goal of this section is to explore the relationship between the borrower's income levels, the loan amount received and default behavior.

The minimum income was **15000** while the maximum was **149999**. The borrowers were then segmented into three income categories: **Low income (15000 to 59999)**, **Medium income (60000 to 104999)**, **High income (105000 to 149999)**.

Using this columns loan amounts were then aggregated by both income category and employment type.

4. CREDIT SCORE ANALYSIS

To analyze how credit scores relate to loan amounts, borrowers were grouped into four categories based on their credit score values: **Very Low (200–361)**, **Low (362–524)**, **Medium (525–687)**, and **High (688–849)**. The median loan amount was then calculated for each credit score category to identify borrowing patterns.

5. LOAN TRENDS OVER TIME

Using the loan date field the year column was extracted. The loan amounts issued were then aggregated by year.

6. LOAN AMOUNT CLASSIFICATION

To further understand the borrowers, loan amounts were analyzed by age groups and purpose.

6.1 LOAN AMOUNT BY AGE GROUP

Borrowers were segmented into age-based categories using defined bins: **Teen (18–25)**, **Adults (26–45)**, **Middle Age Adults (46–60)** and **Senior Citizens (61–70)**. The average loan amount for each group was calculated and rounded to two decimal places.

6.2 LOAN AMOUNT BY PURPOSE

Loan amounts were aggregated based on the stated purpose of each loan.

6.3 LOAN AMOUNT BY EMPLOYMENT TYPE

This classification helps understand the number of loans distributed across full-time, part-time, self-employed, and unemployed borrowers, and can inform lending strategies based on occupational risk profiles.

7. DEFAULT ANALYSIS

7.1 TOTAL DEFAULTS

A total of **29,653 loans** were identified as defaulted out of **255,347 records**.

7.2 DEFAULT RATE BY YEAR

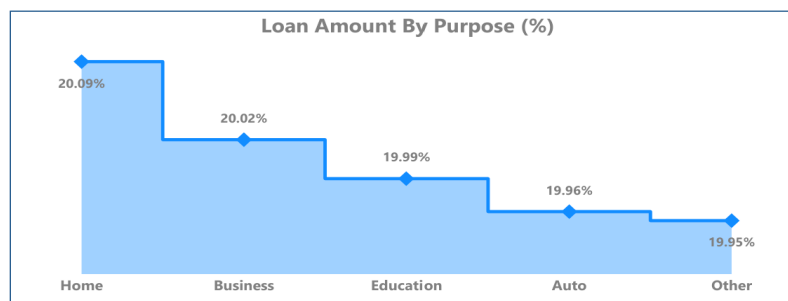
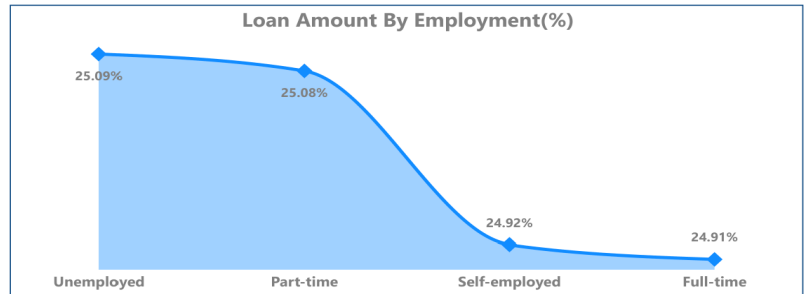
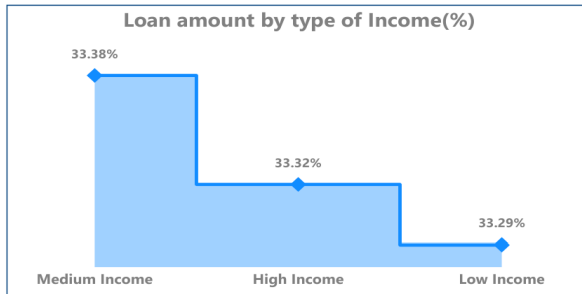
Loan records were grouped by year to calculate annual default rates.

7.3 DEFAULT RATE BY TYPE OF EMPLOYMENT

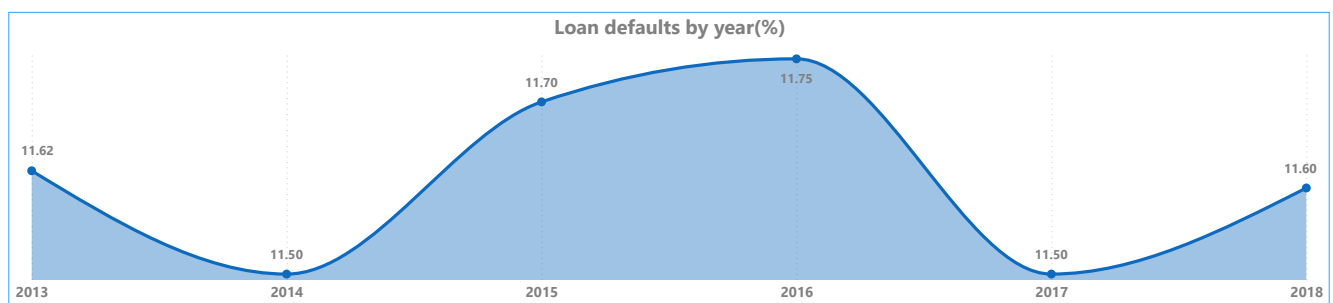
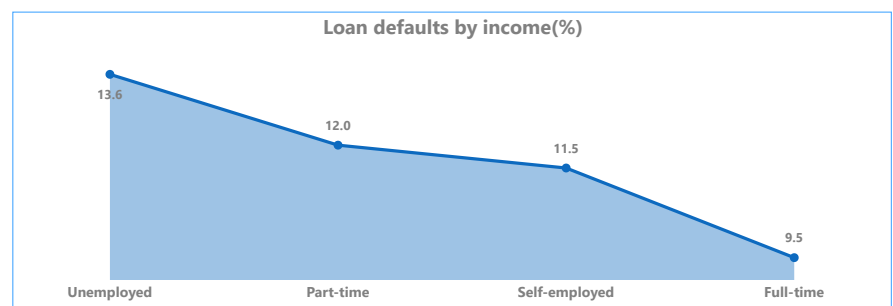
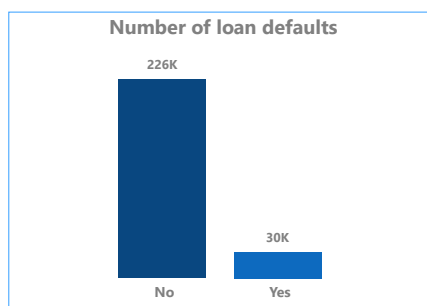
Loans were grouped by employment type to assess how job category influences default risk.

8. VISUALIZATIONS

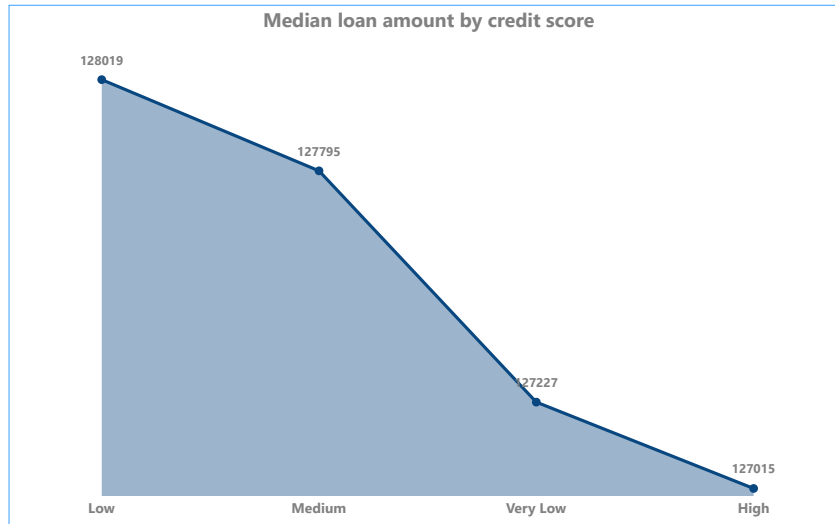
Loan amount classification



Default rate analysis



Median loan amount by credit score



9. KEY INSIGHTS

- The **unemployed and medium** income earners received the largest share of loans disbursed.
- **Home loans** attracted the most borrowers in the market.
- Analysis of **Loan defaults** showed that loans given in **2016** had the highest default rate at **11.75%**, which is only slightly higher than the mean default rate of **11.61%**.
- As one might have expected the rate of default by the **unemployed** is highest at **13.55%** while loans issued to **full time** employees performed best and had a default rate of **9.46%**.
- The overall rate of default being **11.61%** shows a good performance on most of the loans issued.

10. CONCLUSION

10.1. SUMMARY OF FINDINGS

- There should be an increase in the amount and number of business loans provided, specifically **business loans**. Business loans performed second best overall compared to other loan types. There is room for growth and great return here since the **self employed** had the second lowest default rate at **11.5%**. A business loan to the self employed is therefore a chance for a positive return.
- Loan products should be target at full-time income earners since they have a lower default rate currently.
- Home loans and business loans remain attractive to the borrowers.
- The market shows a **slight rise in default rate** after a period of lower default.

10.2 RECOMMENDATIONS

- Consider offering higher loan amounts to full-time employees.
- Exercise caution as the default rate seems to be rising.
- Credit scores should be given more consideration in the lending process.

11. APPENDIX

A. JUPYTER NOTEBOOK LINK

[Data-Analysis/Projects/Loans/Notebook.ipynb at main · ian-obotey/Data-Analysis](#)

B. NOTES ON ASSUMPTIONS AND LIMITATIONS

This analysis was conducted using tutorial data provided for learning purposes. While the dataset is synthetic the analysis techniques mirror real-world scenarios.